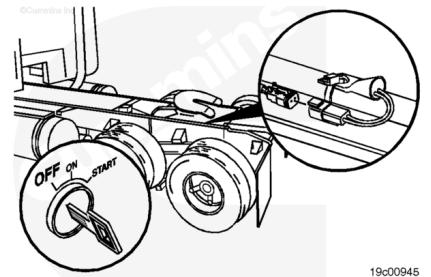


Trainings ▾

Resistance Check

⚠ CAUTION ⚠

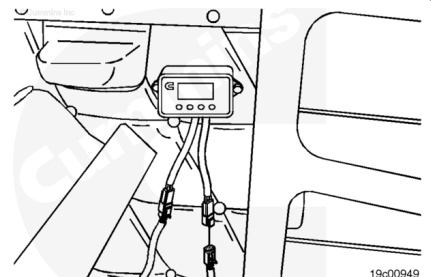
To avoid pin and harness damage, use test lead, Part Number 3822758, when taking a measurement.



Turn the keyswitch to the OFF position. Disconnect the ambient air temperature sensor from the temperature sensor harness.

Set the multimeter to measure resistance.

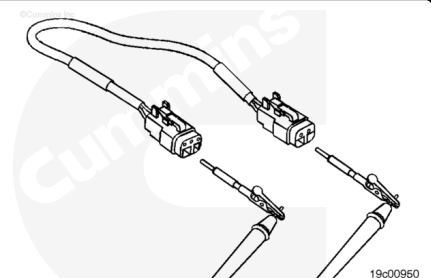
Disconnect the temperature sensor harness from the cab thermostat.



Touch one of the multimeter leads to pin 1 of the temperature sensor harness connector on the ambient air temperature sensor side.

Touch the other multimeter lead to pin 1 of the temperature sensor harness connector, cab thermostat side.

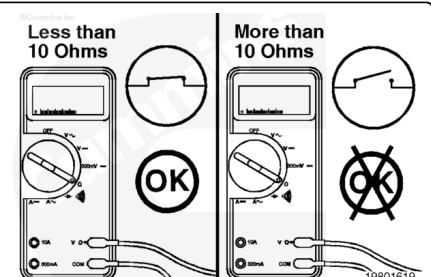
Read the value displayed on the multimeter.



The multimeter **must** display a reading of less than 10 ohms, which is a closed circuit.

If the circuit is **not** closed, repair the defective temperature sensor harness connector or replace the temperature sensor harness. Refer to Procedure 019-202

(/qs3/pubsys2/xml/en/procedures/97/97-019-202.html), 019-



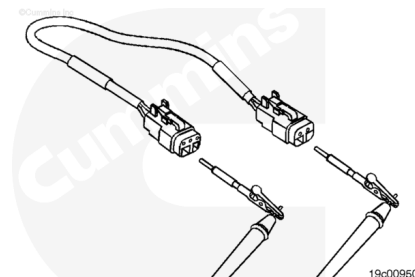
203 (/qs3/pubsys2/xml/en/procedures/97/97-019-203.html), or 019-296 (/qs3/pubsys2/xml/en/procedures/97/97-019-296.html).

Repeat the resistance check for the return wire.

Touch one of the multimeter leads to pin 2 of the temperature sensor harness connector, sensor side. Touch the other multimeter lead to pin 3 of the temperature sensor harness connector, cab thermostat side. Read the value displayed on the multimeter.

The multimeter **must** display a reading of less than 10 ohms. If the circuit is **not** closed, repair the defective sensor harness connector or replace the temperature sensor harness. Refer to Procedure 019-202 (/qs3/pubsys2/xml/en/procedures/97/97-019-202.html), 019-203 (/qs3/pubsys2/xml/en/procedures/97/97-019-203.html), or 019-296 (/qs3/pubsys2/xml/en/procedures/97/97-019-296.html).

Connect all components after completing the repair.



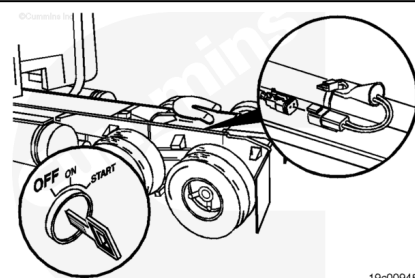
Check for Short Circuit to Ground

⚠ CAUTION ⚠

To avoid pin and harness damage, use test lead, Part Number 3822758, when taking a measurement.

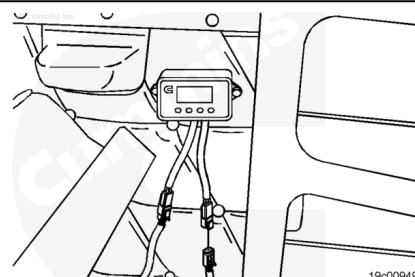
Turn the keyswitch to the OFF position.

Disconnect the ambient air temperature sensor from the temperature sensor harness.



Disconnect the temperature sensor harness from the cab thermostat.

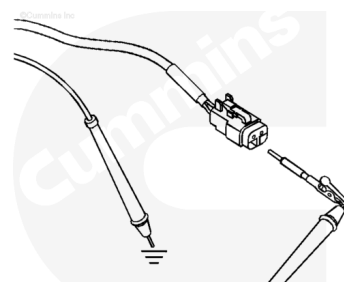
Set the multimeter to measure resistance.



Touch one of the multimeter leads to pin 1 of the temperature sensor harness connector on the ambient air temperature sensor side.

Touch the other multimeter lead to ground.

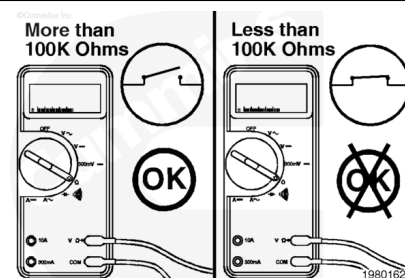
Read the value displayed on the multimeter.



19c00951

The multimeter **must** display more than 100k ohms, which is an open circuit. If the circuit is **not** open, there is a short circuit to ground.

Repair or replace the temperature sensor harness. Refer to Procedure 019-202 (</qs3/pubsys2/xml/en/procedures/97/97-019-202.html>) or 019-296 (</qs3/pubsys2/xml/en/procedures/97/97-019-296.html>).



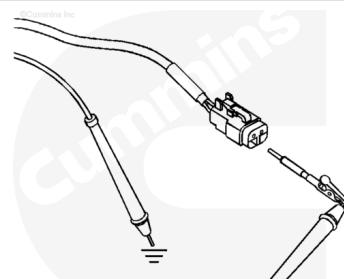
19801621

Touch one of the multimeter leads to pin 2 of the temperature sensor harness connector on the ambient air temperature sensor side. Touch the other multimeter lead to ground.

Read the value displayed on the multimeter.

The multimeter **must** display more than 100k ohms (open circuit). If the circuit is **not** open, there is a short circuit to ground. Repair or replace the temperature sensor harness. Refer to Procedure 019-202 (</qs3/pubsys2/xml/en/procedures/97/97-019-202.html>) or 019-296 (</qs3/pubsys2/xml/en/procedures/97/97-019-296.html>).

Connect all components after completing the repair.



19c00951

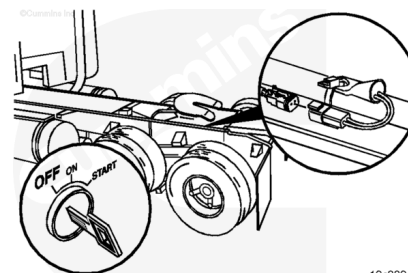
Check for Short Circuit from Pin to Pin

⚠ CAUTION ⚠

To avoid pin and harness damage, use test lead, Part Number 3822758, when taking a measurement.

Turn the keyswitch to the OFF position.

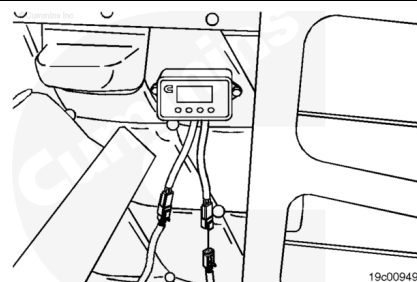
Disconnect the ambient air temperature sensor from the temperature sensor harness.



19c00945

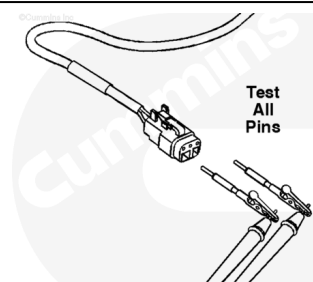
Disconnect the temperature sensor harness from the cab thermostat.

Set the multimeter to measure resistance.



Touch one of the multimeter leads to pin 1 of the temperature sensor harness connector, cab thermostat side. Touch the other multimeter lead to each of the other three pins in the connector. Repeat the pin-to-pin check for the other 3 pins in the connector.

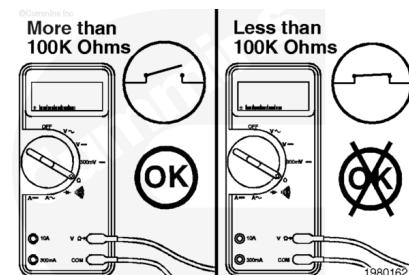
Read the value displayed on the multimeter.



The multimeter **must** display more than 100k ohms, which is an open circuit. If the circuit is **not** open, there is a short circuit between the two pins.

Repair or replace the temperature sensor harness. Refer to Procedure 019-203 (</qs3/pubsys2/xml/en/procedures/97/97-019-203.html>) or 019-296 (</qs3/pubsys2/xml/en/procedures/97/97-019-296.html>).

Connect all components after completing the repair.



Check for Short Circuit to External Voltage Source

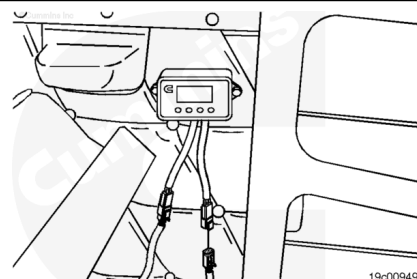
⚠ CAUTION ⚠

To avoid pin and harness damage, use test lead, Part Number 3822758, when taking a measurement.

Disconnect the temperature sensor harness from the cab thermostat.

Set the multimeter to measure VDC.

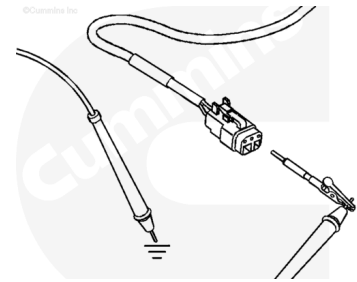
Turn the keyswitch to the ON position.



Touch one of the multimeter leads to pin 1 of temperature sensor harness connector on the cab thermostat side.

Touch the other multimeter lead to ground.

Read the value displayed on the multimeter.

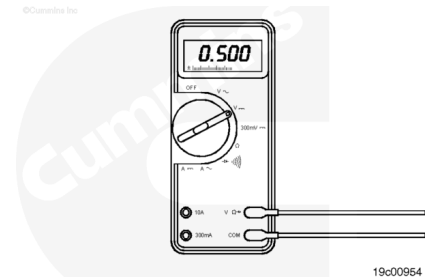


19c00963

The multimeter **must** display less than 0.5 VDC. If the circuit is **not** less than 0.5 VDC, there is a short circuit to an external voltage source.

Remove the external voltage source.

Note : An external voltage source is any wire in the temperature sensor harness that carries voltage.



19c00964

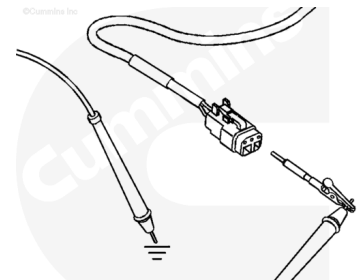
Repeat the check for the return wire.

Touch one of the multimeter leads to pin 3 of temperature sensor harness connector on the cab thermostat side. Touch the other multimeter lead to ground.

Read the value displayed on the multimeter.

The multimeter **must** display less than 0.5 VDC. If the circuit is **not** less than 0.5 VDC, there is a short circuit to an external voltage source. Remove the external voltage source.

Connect all components after completing the repair.



19c00963